



3 Make this layer just as you did in step 2. Draw up the dishwashing liquid with the baster and trickle it in slowly against the side of the glass. Dishwashing liquid is water with large detergent molecules dissolved in it.



4 You're now ready to add the fourth layer, which is water. You can make this any color you like by adding a few drops of food coloring. Remember to trickle the liquid slowly! Water molecules are very small and tightly packed.



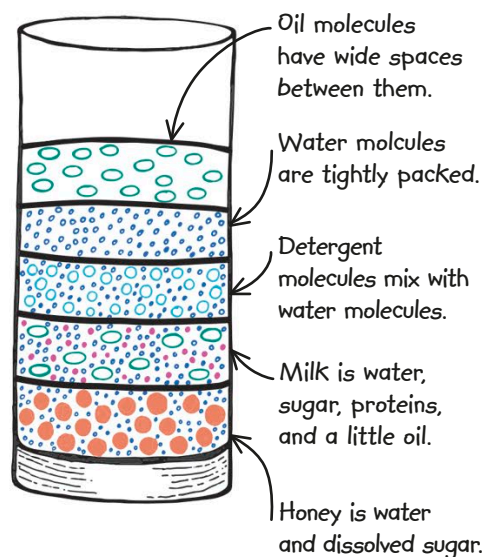
5 Finally, add vegetable oil, although olive oil also works. Interestingly, if you had put the oil in first instead of last, it would still have risen to the top because it's the least dense liquid—but you would spoil your tower!



6 Now gently drop in small objects, like a bolt, tomato, and ping-pong ball. The bolt sinks to the bottom because it is denser than the honey. The tomato sinks until it meets the milk. What happens to the ping-pong ball?

HOW IT WORKS

Although water molecules crowd closely together, each one has low mass, so water's density is fairly low. When substances dissolve in water, their molecules sneak in between the water molecules, increasing the solution's density. Oil molecules are bigger and do not pack together so tightly, which means it has a low density.



REAL WORLD SCIENCE

OIL SPILL



Occasionally, tanker ships carrying oil accidentally spill their contents. This is bad for sea life, and an oil spill is difficult to clean up. What makes the task a bit easier is the fact that oil floats on water. The oil can be scooped up out of the water or sprayed with detergents, which helps to dissolve it.